

Acoustic Biodiversity Mapping

A global location-level biodiversity metric from Xeno-canto recordings

Generated 2026-09-08

OBJECTIVE

Derive a single metric that classifies a geographic location as biodiversity 'good' or 'bad', using a global archive of wildlife sound recordings and their metadata.

DATA

- 759,767 recordings across 5 continents (Xeno-canto), each with 42 acoustic indices computed by a SLURM batch pipeline (compute_indice.py over Butterworth-filtered audio).
- Metadata flattened from 766,747 API records (species, coordinates, quality, device).
- Acoustic scores joined to metadata on recording id (>99.99% match).

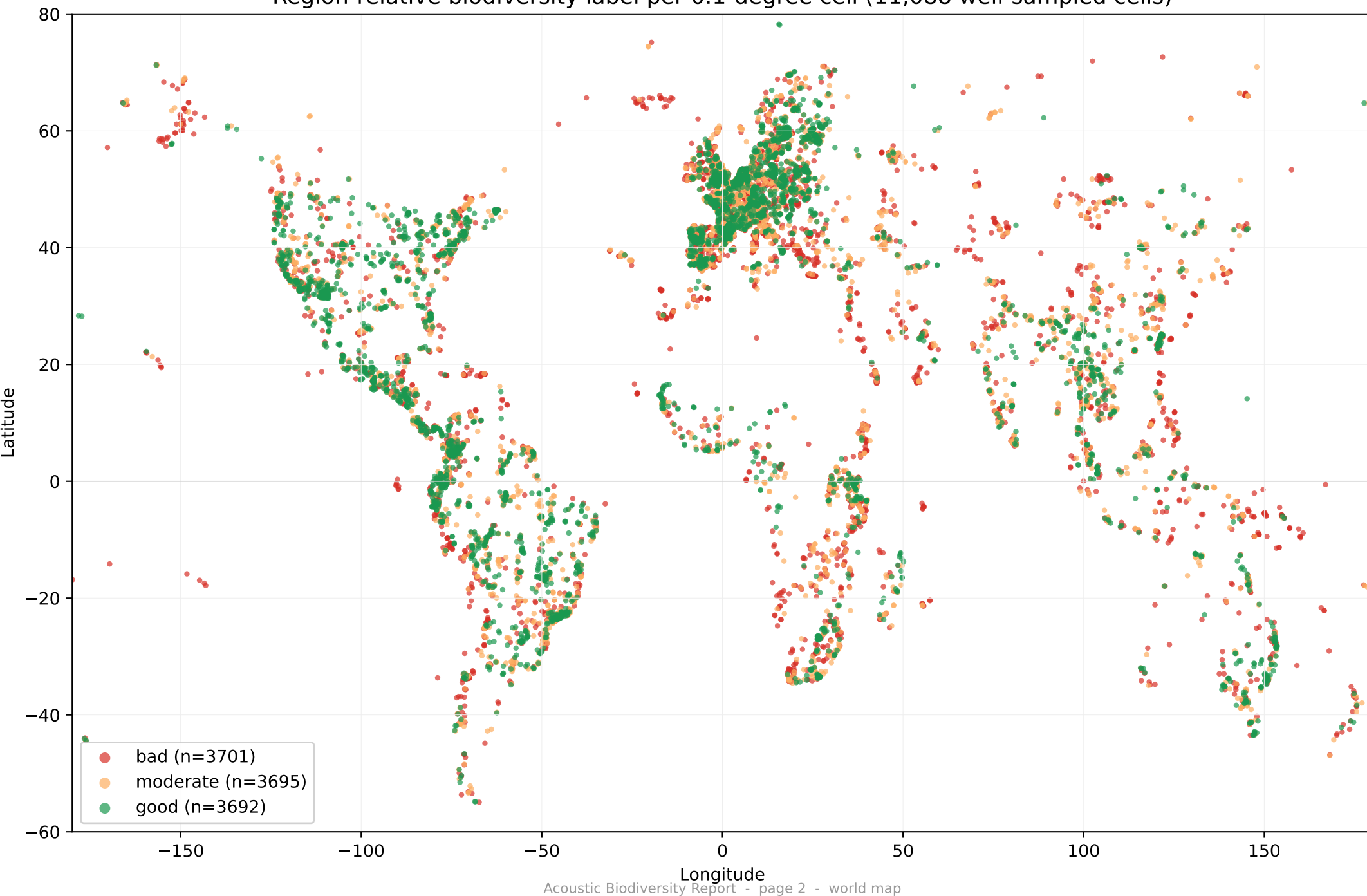
METHOD

1. Aggregate all recordings into 0.1-degree (~11 km) grid cells -> 40,914 cells.
2. Per cell, compute species richness from recorded species (genus+species and the 'also' co-occurring-species field), corrected for sampling effort by Hurlbert rarefaction to 10 recordings (S_{rare10}).
3. Test whether the acoustic indices predict richness (validation against ground truth).
4. Label each well-sampled cell good / moderate / bad relative to its 10-degree latitude band (region-relative tertiles).

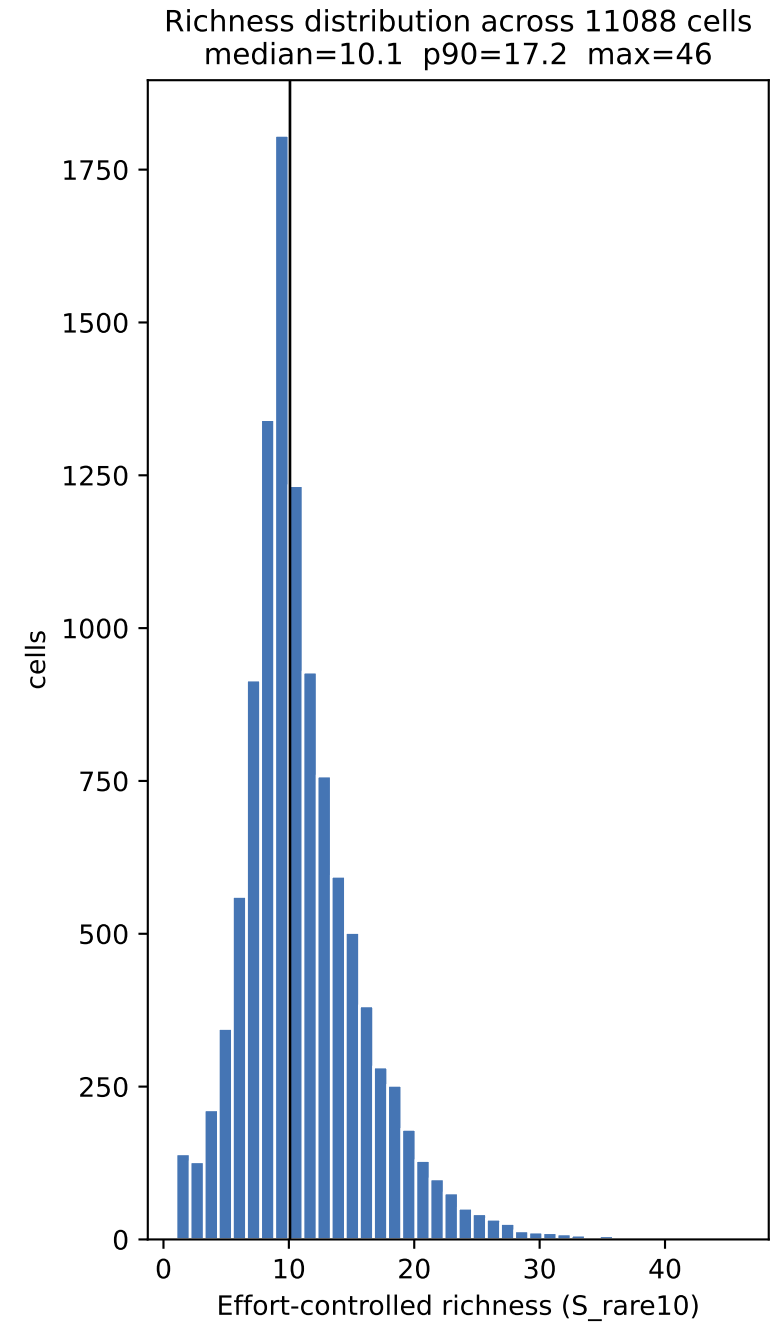
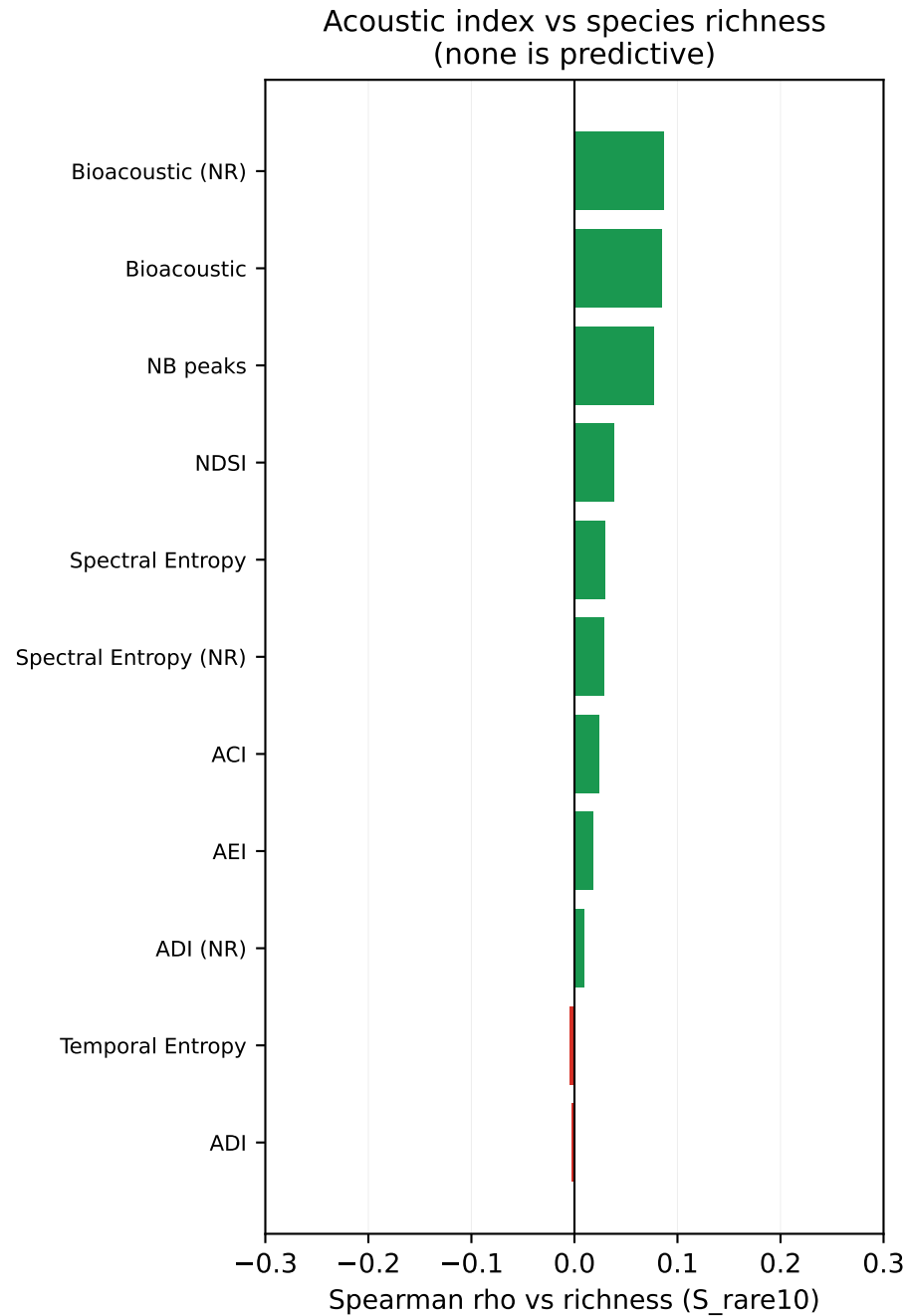
HEADLINE RESULT

- 11,088 cells had enough recordings (≥ 10) to score confidently.
- The acoustic indices did NOT predict species richness (best |Spearman rho| ~ 0.09). The defensible biodiversity metric is therefore effort-controlled species richness, not any acoustic index -- the indices are unreliable on targeted (non-soundscape) recordings that dominate the archive.
- Final labels: 3692 good, 3695 moderate, 3701 bad.

Region-relative biodiversity label per 0.1-degree cell (11,088 well-sampled cells)



Validation: does the audio carry the biodiversity signal?



The Metric & Results

METRIC DEFINITION

location = 0.1 deg x 0.1 deg grid cell (~11 km)
biodiversity value = S_{rare10} , the expected number of distinct recorded species in 10 recordings (Hurlbert sample-based rarefaction; controls for recording effort)
label = tertile of S_{rare10} WITHIN the cell's 10-deg latitude band: good = top third, bad = bottom third
scope = cells with ≥ 10 recordings (11,088 of 40,914 cells)

GLOBAL VALUE RANGE (S_{rare10})

min 1.0 p25 8.2 median 10.1 p75 13.3 p90 17.2 max 46

Median richness by continent (well-sampled cells)

Continent	Median richness	Scored cells
america	10.9	3415
europa	10.2	4643
australia	9.9	404
africa	9.7	1003
asia	9.3	1623

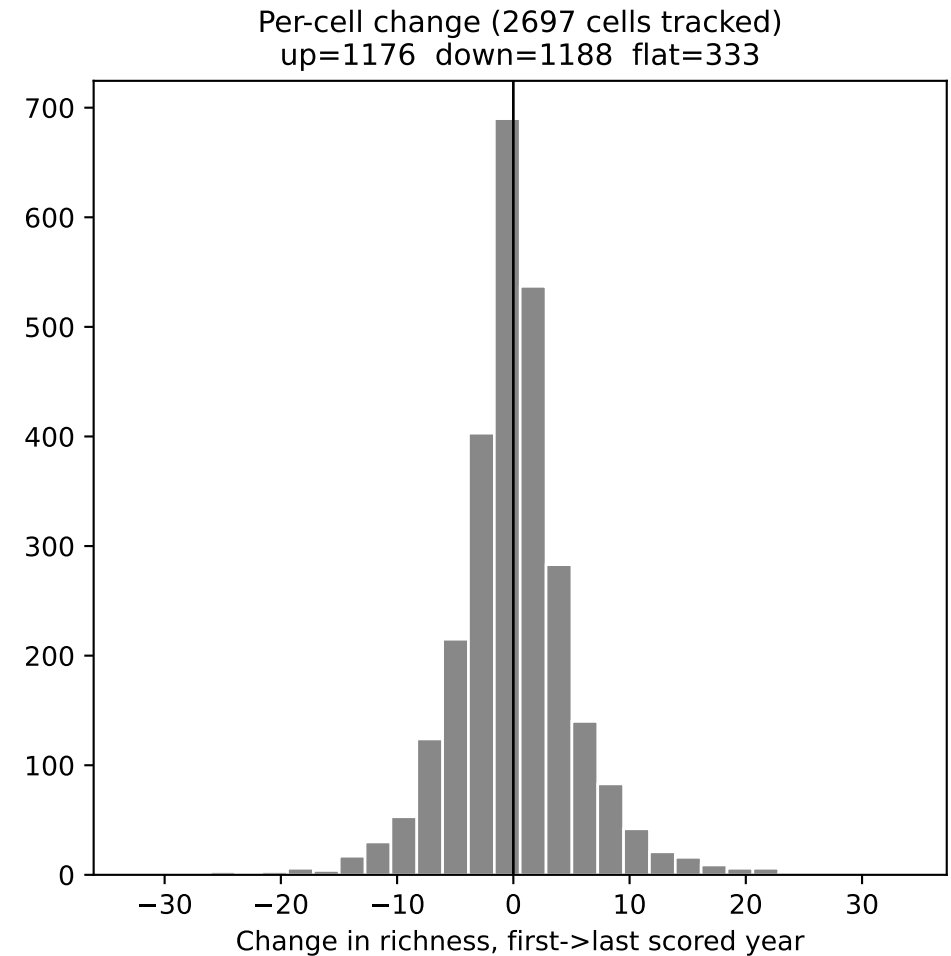
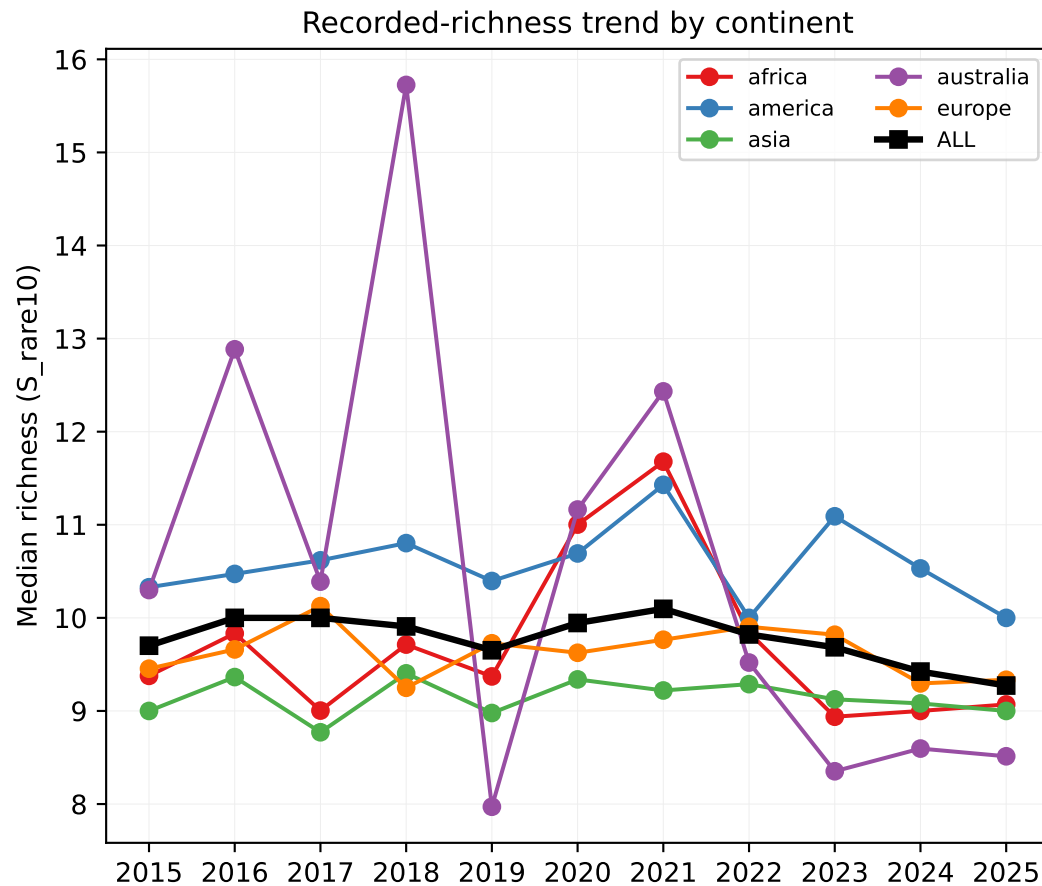
KEY LIMITATIONS (must accompany any use of this metric)

1. Acoustic indices are non-predictive here. ACI/ADI/NDISI etc. were built for passive soundscape monitoring; 69% of archive clips are single-target recordings (median 24 s), so per-clip indices do not reflect site diversity.
2. 'Richness' = RECORDED species, not true species. It reflects recordist effort and interest. Rarefaction controls sample SIZE but not observer bias.
3. Weak latitude gradient. Effort-controlled richness is nearly flat across latitude (thresholds ~10-13 everywhere) -- evidence the signal is effort-driven, not the true tropical biodiversity peak. Region-relative labelling is used so scores mean 'rich for its region'.
4. Coverage. Only 27% of cells are confidently scored; the rest are 'insufficient_data' (< 10 recordings).

OUTPUT FILES

grid_cells.csv	- per-cell features + richness (all 40,914 cells)
grid_cells_labeled_regional.csv	- region-relative good/moderate/bad labels
biodiversity_map.csv	- lat, lon, label, code -> ready for GIS/folium

Year-by-year (2015-2025): temporal slices of the metric



COVERAGE: 14,487 scored cell-years; 2697 cells scored in ≥ 2 years, 10 in all 11.

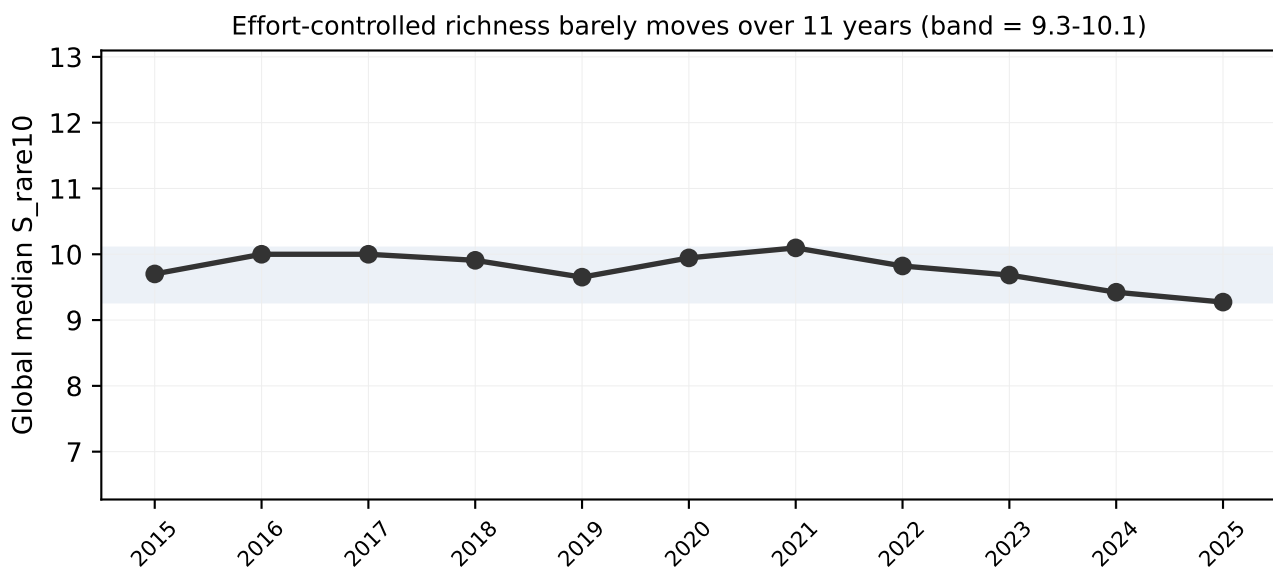
CAUTION: even an 11-year window cannot show real biodiversity change. These movements reflect WHICH cells were recorded and by WHOM each year (effort + observer turnover), not ecological gain or loss. The global change (9.7 $\rightarrow$ 9.3) tracks recording effort, not nature. Use year slices to study sampling coverage over time -- not as a biodiversity time series.

OUTPUTS: grid_cells_yearly.csv (per cell-year), cell_change.csv (per tracked cell).

Ten-Year Findings (2015-2025)

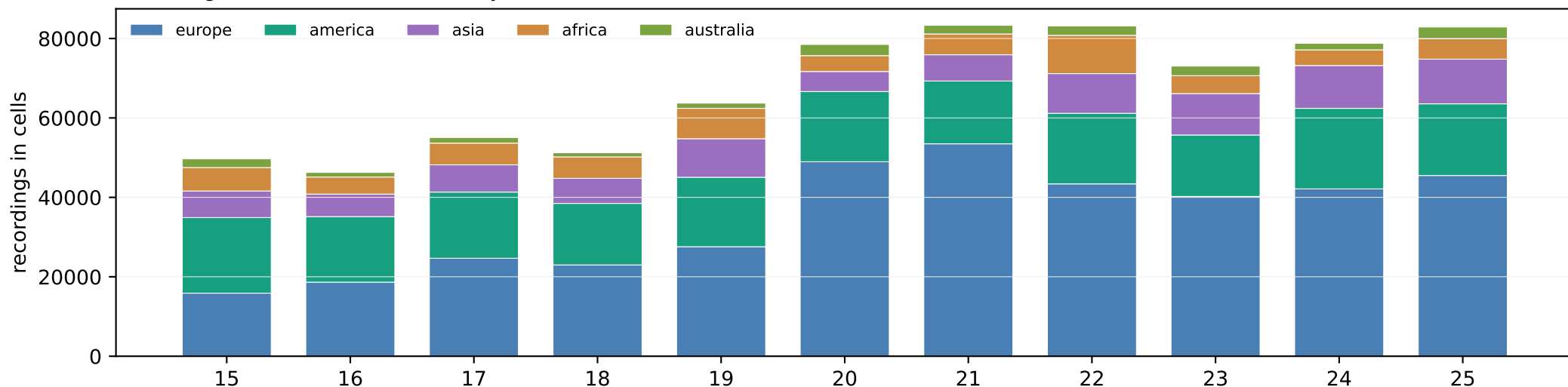
What the decade-scale data adds to the analysis

1. MORE DATA MADE THE ACOUSTIC INDICES LOOK WORSE, NOT BETTER.
Expanding from the 2023-2025 subset to the full 2015-2025 archive (759,767 recordings, 11,088 scored cells) shrank the best index correlation from $|\rho| \sim 0.16$ to ~ 0.09 . A real signal sharpens with more data; this faded toward zero -- the strongest evidence yet that the indices carry no site-biodiversity signal in this archive.
2. A DECADE OF EFFORT-CONTROLLED RICHNESS IS ESSENTIALLY FLAT.
Global median S_{rare10} stays within 9.3-10.1 across all 11 years (2015->2025: 9.7 -> 9.3). This reframes the metric as a sampling-effort measure, not ecology. The only structure is a mild recent dip that tracks the incompleteness of recent uploads.
3. TEMPORAL CHANGE IS A COIN-FLIP (RANDOM-WALK SIGNATURE).
Of 2,697 cells tracked across ≥ 2 years, 1,176 rose and 1,188 fell (flat=333). That near-perfect symmetry over a decade is the fingerprint of observer turnover and noise, not directional change.
4. SPATIALLY DECADE-RICH, BUT TEMPORALLY STILL STARVED.
Coverage grew to 40,914 cells / 11,088 scored -- a much denser map -- yet only 10 cells were recorded in all 11 years (2,697 in ≥ 2). The biodiversity MAP is far stronger; a true TIME SERIES remains impossible from this data.

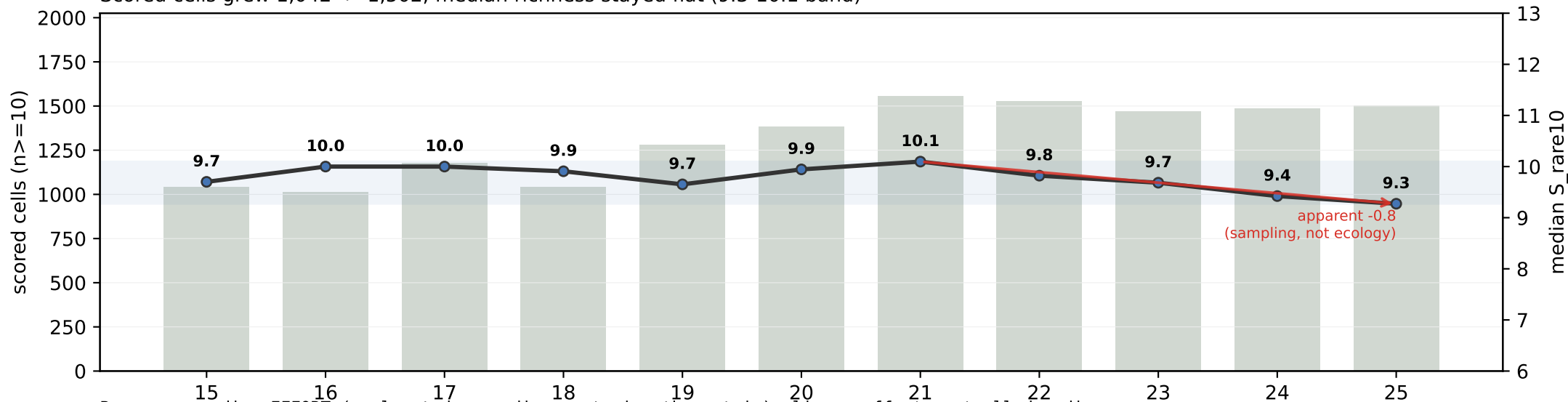


Year-by-year: effort vs. the metric (2015-2025)

Recording effort behind the metric, by continent (49,684 -> 82,884, +66%)



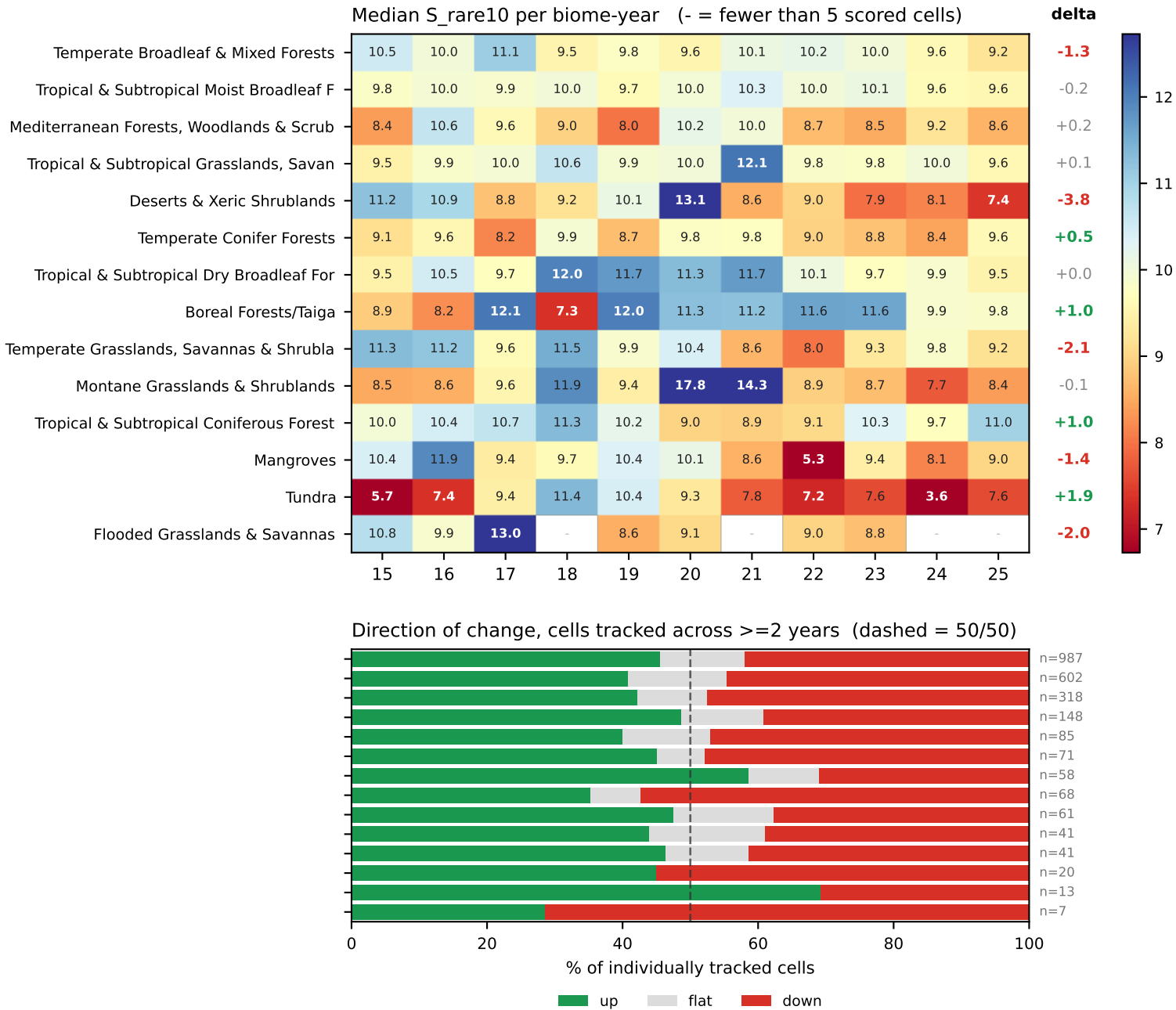
Scored cells grew 1,042 -> 1,502; median richness stayed flat (9.3-10.1 band)



Bars = recording EFFORT (geolocated recordings entering the metric); line = effort-controlled median richness. Effort rose ~70% over the decade while the metric stayed inside a narrow band. The apparent recent dip (red arrow) is NOT ecological degradation: it tracks the lower completeness of recent uploads and shifting recordist coverage. Do not read the decline as biodiversity loss.

Year-by-Year Change by Biome (2015-2025)

Effort-controlled richness (*S_rare10*) split by terrestrial biome



NO BIOME SHOWS COHERENT DIRECTIONAL CHANGE.

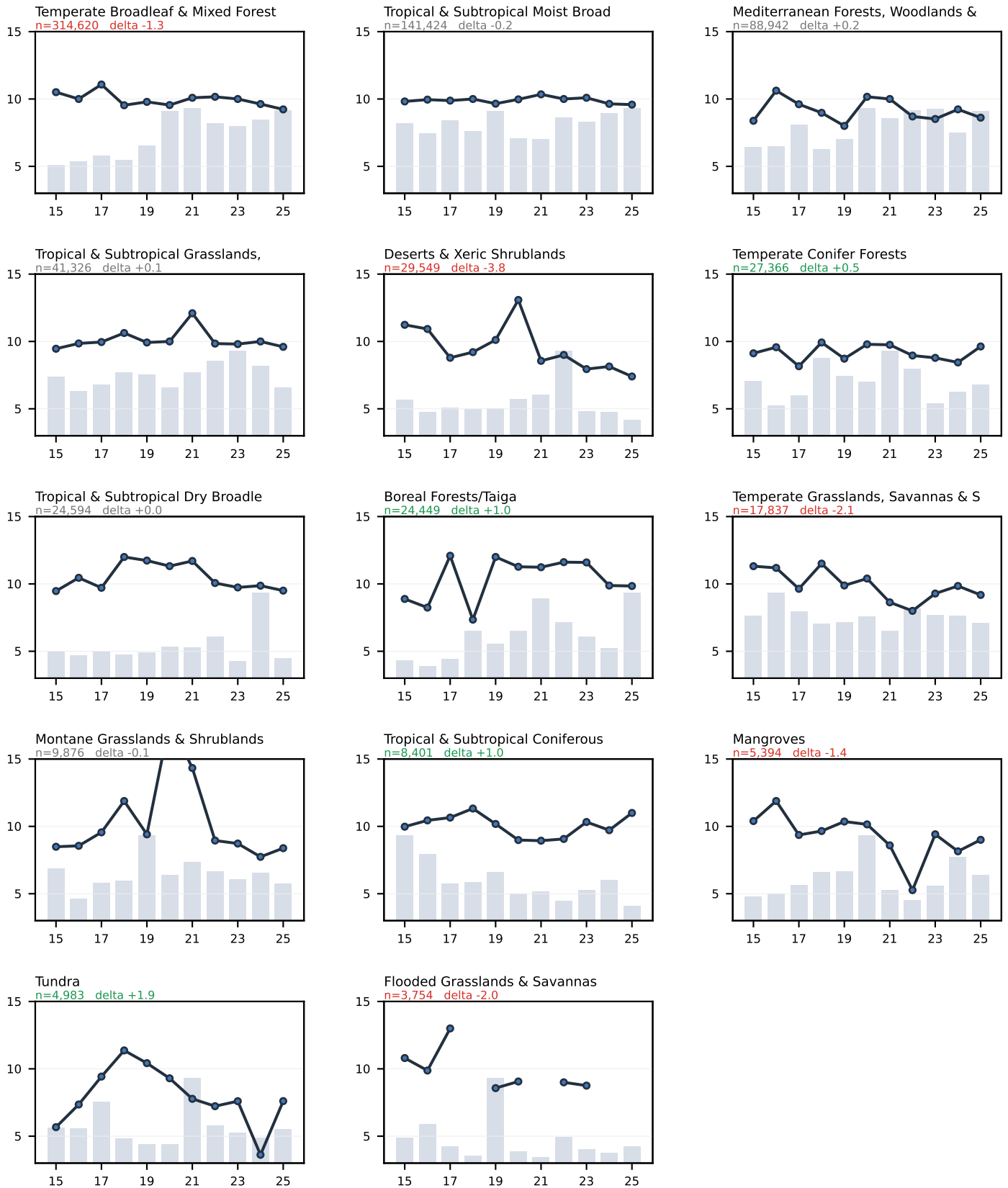
Every biome's year-to-year jitter is as large as its 2015->2025 delta, and the cells tracked inside each biome split close to 50/50 (1,111 up vs 1,099 down overall). The largest delta -- Deserts & Xeric Shrublands at -3.8 -- rests on a series that swings by more than that between adjacent years.

READ THIS AS SAMPLING, NOT ECOLOGY. *S_rare10* controls for the NUMBER of recordings in a cell, but not for who recorded, for how long, or with what target. A biome's series therefore moves as its recordist community turns over. The biomes with the fewest recordings -- Tundra, Mangroves, Flooded Grasslands -- swing hardest, which is the signature of small samples rather than of habitat change.

METHOD. Each 0.1-degree cell takes the modal biome of the recordings inside it. 99.4% of recordings fall in their cell's modal biome and only 1.6% of cells span more than one, so boundary cells cannot drive these patterns. Biomes are ordered by recording volume.

Each Biome, Year by Year (2015-2025)

Line = median S_{rare10} (left axis). Bars = recordings entering the metric (right axis).

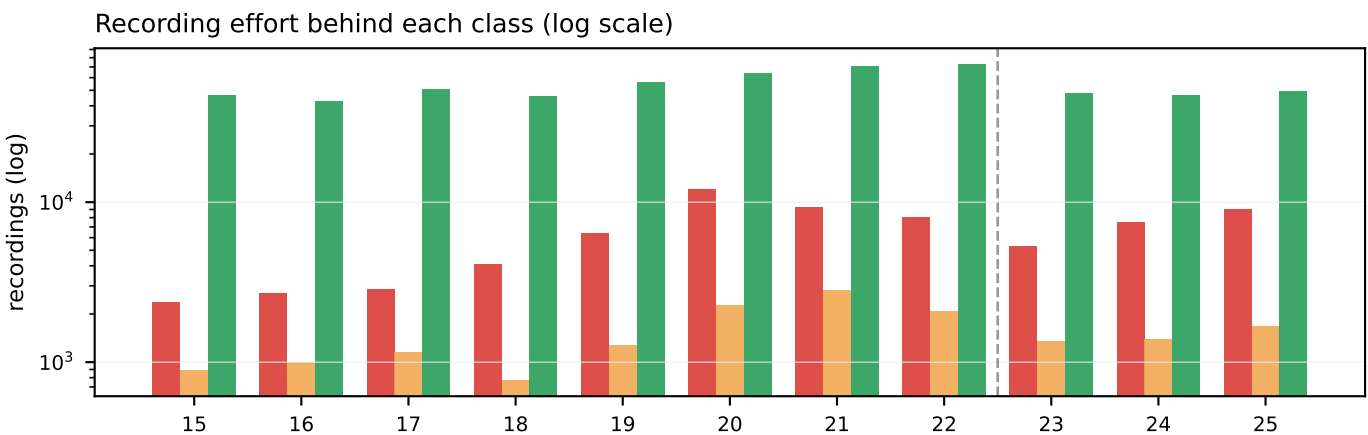
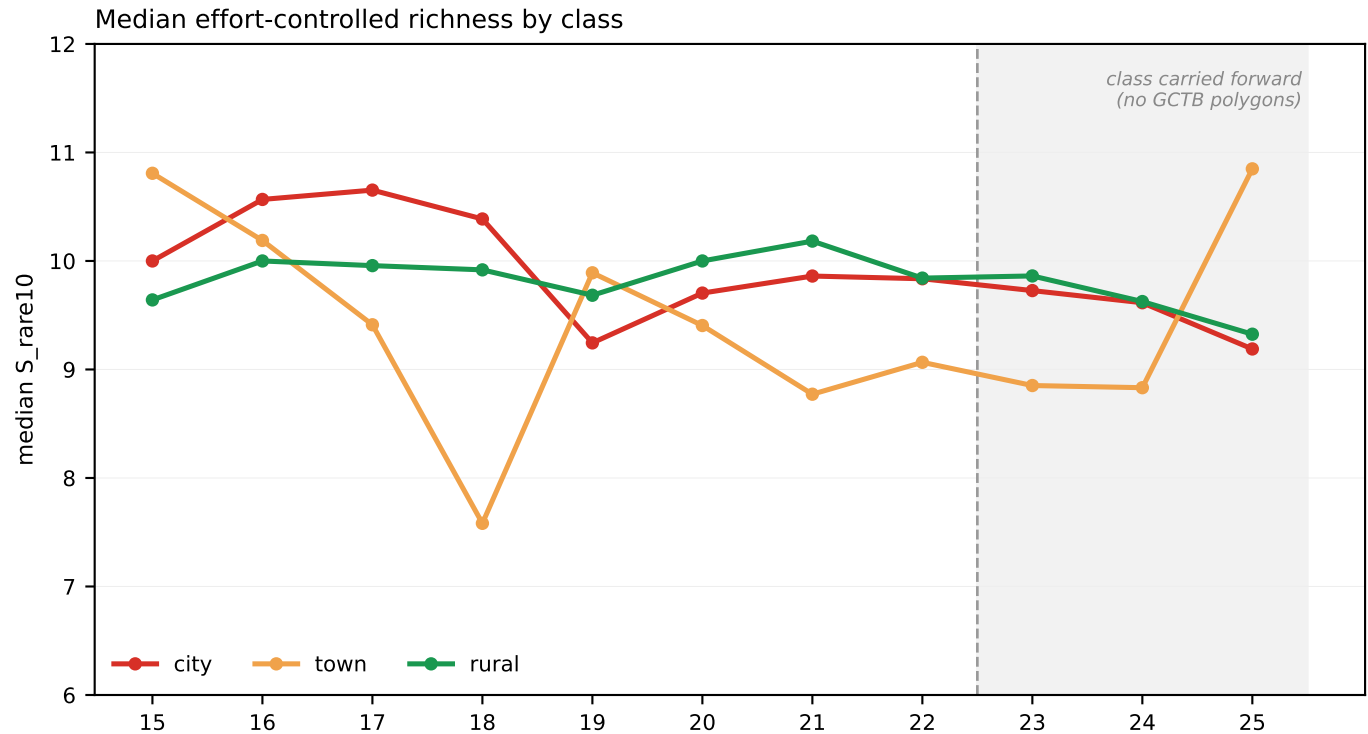


Every panel is drawn on the SAME richness axis (3-15), so the series can be compared directly. Read the grey bars first: where effort is small the line is jagged, and where effort is large it is flat. That is the whole pattern -- Tundra, Mangroves and Flooded Grasslands swing by several species between adjacent years on a few hundred recordings, while Temperate Broadleaf & Mixed Forests, with 314,620, barely moves.

No panel shows a monotonic decade-long trend. The deltas printed above each panel are first-vs-last-year differences, not fitted trends, and in every biome the year-to-year jitter is at least as large as the delta -- so none of them is distinguishable from noise.

Year-by-Year Change by Urban Class (2015-2025)

Effort-controlled richness (S_{rare10}) for city / town / rural cells



NEITHER CITY NOR RURAL RICHNESS TRENDS.

Over the labelled window (2015-2022) the city median moves -0.2 and rural $+0.2$. Both are far smaller than the year-to-year jitter in either series, so neither is distinguishable from noise. The town series (only 719 cells) swings by ~ 3 species between adjacent years, which is what a small sample looks like.

THE TRACKED CELLS SAY NO CHANGE. Cells followed across ≥ 2 years split city 109/103 up/down and rural 945/945 -- rural is an exact coin flip. A real divergence between urban and rural biodiversity would show up here first, and it does not.

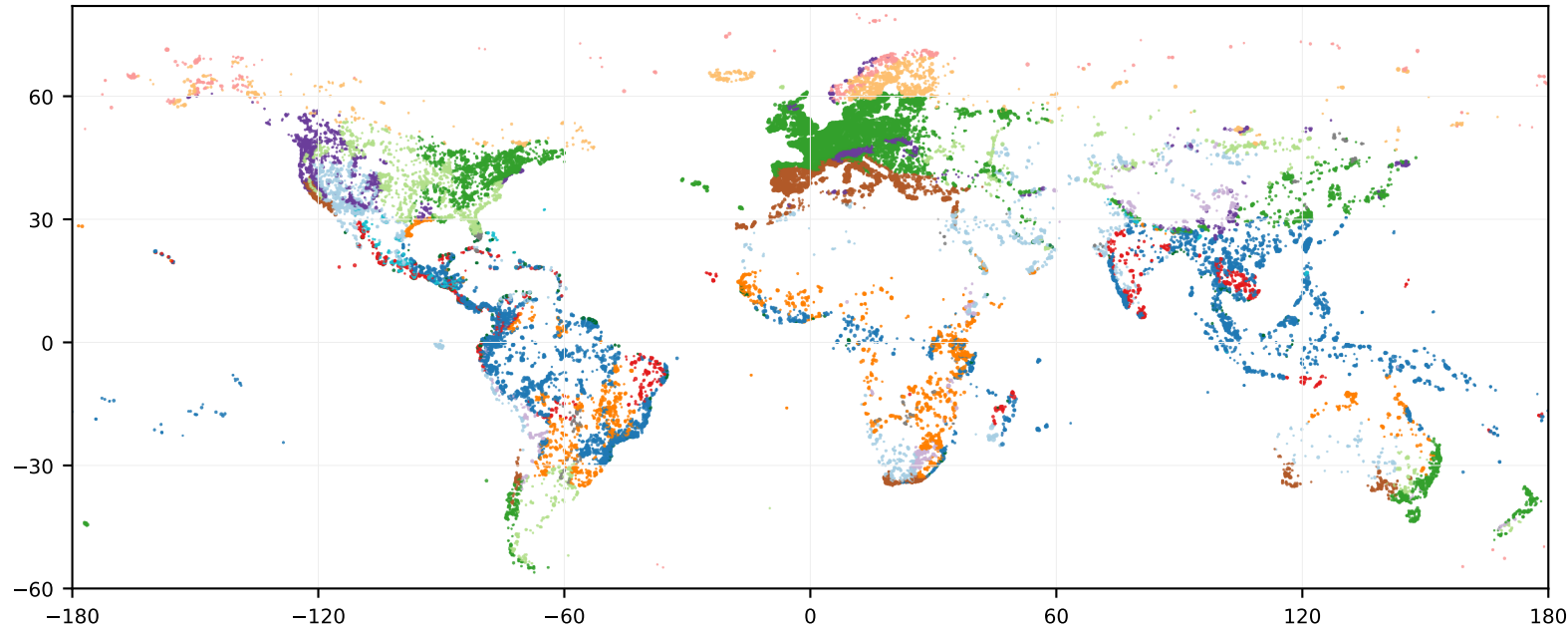
COVERAGE CAVEAT. GCTB built-up polygons stop at 2022, so only 510,923 recordings (2015-2022) carry a real class. Urban class is a property of the PLACE, so each cell's class is carried forward to 2023-2025 (shaded). Those three years rest on an assumption -- that cells did not change built-up status -- not on measurement.

Cells take the modal class of their recordings: 97.0% of recordings fall in their cell's modal class and 5.8% of cells are mixed (higher than the 1.6% for biomes, because city boundaries cut through 0.1-degree cells far more often than biomes do).

Recording Coverage by Biome

40,668 grid cells (0.1 deg), each coloured by the modal biome of its recordings

Dot size = recordings in that cell (log scale)



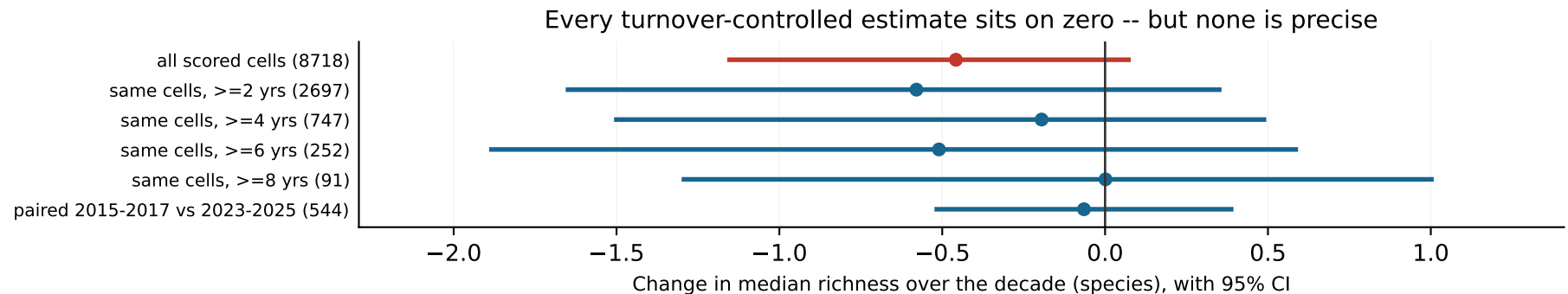
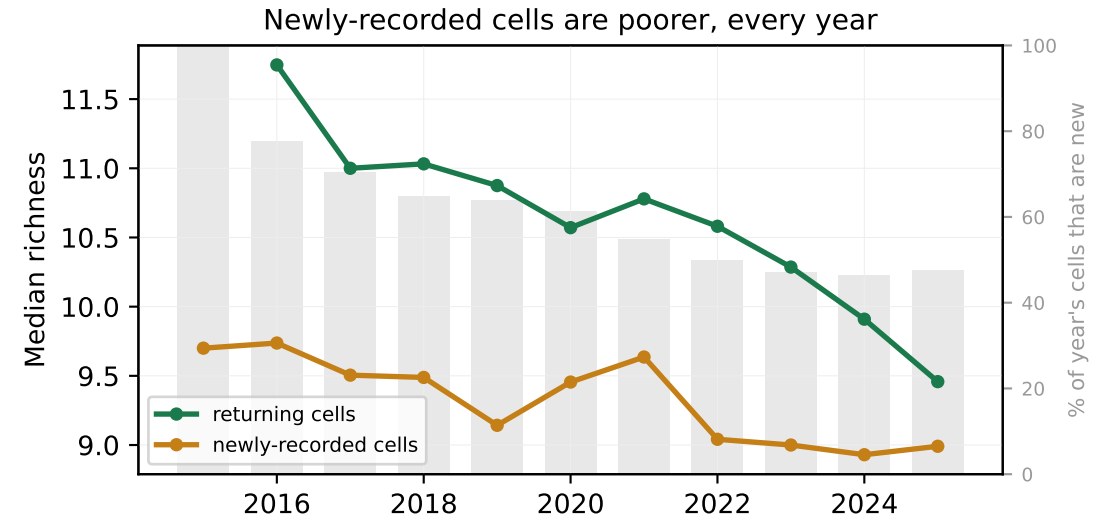
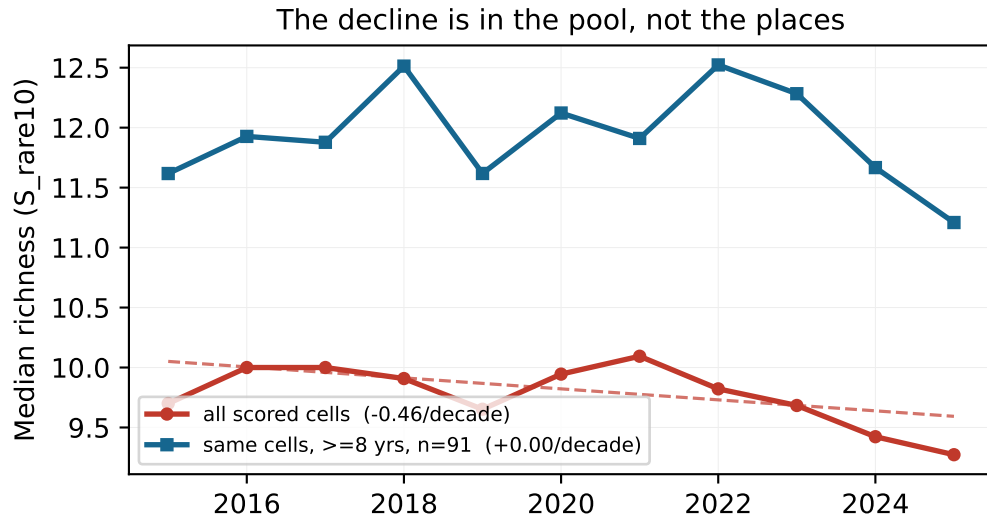
biome (cells / recordings)

- Temperate Broadleaf & Mixed Forests
14,065 cells / 313,586 rec
- Tropical & Subtropical Moist Broadleaf
7,034 cells / 141,420 rec
- Mediterranean Forests, Woodlands & Scr
4,246 cells / 88,942 rec
- Tropical & Subtropical Grasslands, Sav
2,581 cells / 41,326 rec
- Deserts & Xeric Shrublands
2,427 cells / 29,549 rec
- Temperate Conifer Forests
2,671 cells / 27,365 rec
- Tropical & Subtropical Dry Broadleaf F
1,312 cells / 24,594 rec
- Boreal Forests/Taiga
1,687 cells / 24,449 rec
- Temperate Grasslands, Savannas & Shrub
2,179 cells / 17,837 rec
- Montane Grasslands & Shrublands
823 cells / 9,875 rec
- Tropical & Subtropical Coniferous Fore
445 cells / 8,401 rec
- Mangroves
365 cells / 5,394 rec
- Tundra
574 cells / 4,981 rec
- Flooded Grasslands & Savannas
259 cells / 3,754 rec

COVERAGE IS NOT ECOLOGICAL COVERAGE. The map shows where recordists go, not where biomes are. Europe is saturated across a single biome band while whole tropical biomes are covered by scattered cells: Temperate Broadleaf & Mixed Forests holds 14,065 cells, the three smallest together fewer than 1,198. This is the sampling bias behind every per-biome number here -- a biome comparison is a comparison of recordist communities as much as of habitats, and the biomes with fewest cells are exactly those whose yearly series swing hardest on pages 8 and 9.

Cells take the modal biome of their recordings; 634 (1.6%) contain more than one and are drawn in their majority colour. Polygons: RESOLVE Ecoregions 2017 (Dinerstein et al., CC-BY 4.0). Colours are categorical only.

Is there a ten-year trend? Turnover, not change



NO. Raw, the series falls 9.70 $\rightarrow$ 9.27 (Theil-Sen -0.46 species/decade, Kendall tau -0.44, $p = 0.061$) -- marginal, and not significant. Follow the SAME cells and it disappears: the ≥ 8 -year panel gives +0.00/decade and the 544 paired cells -0.065 (269 up / 275 down).

MECHANISM: 46-78% of each year's scored cells were never recorded before, and newly-recorded cells sit below returning ones in 10 of 10 years. The archive keeps expanding into thinner locations, which drags the pooled median down while no individual place changes.

LIMIT: the controlled estimates are UNDERPOWERED, not proof of stability. The paired CI is [-0.52, +0.39] and the ≥ 8 -year panel spans more than two species -- a real decline of up to ~ 0.5 species/decade would not be detectable here. Read this as no evidence of a trend, plus a well-identified artefact that explains the apparent one.

CONFOUND: the steepest fall is the last three years, exactly where the data source changes from the historical backfill to the base+gap run. Year and processing path cannot be separated with what is on disk, so even the marginal raw decline is suspect.